

Absorptive But Not Anechoic

Simulating an anechoic chamber before you trust it: what the models predict, which failures they catch, and what it costs to find out. Acoustic scope only.

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The result everything else hangs off. In 2019 Anthony Nash qualified a 51 m³ commercially built anechoic chamber against ISO 3745 Annex A. On pink noise in one-third-octave bands it **passed**; on pure tones at the same band centres it **failed** — a 2056 Hz tone fitted a decay of −26.4 dB per decade of distance, about 8 dB per doubling, which is not physically possible in a free field and is explicable only as a reflection cancelling the direct wave.

Why it matters here. Band noise averages over the interference pattern; a tone sits inside it, and drone noise is tonal. A chamber qualified with noise is not qualified for this rig — and a simulation excited with noise reproduces the passing answer while the room is failing.

In short, the chamber is absorptive but not anechoic.

A. Nash, *Qualification of an anechoic chamber*, ICA 2019 · on a chamber with foam wedges on all six surfaces, an expanded-metal floor grate, and no evidence its wedges were ever impedance-tube tested

“Simulating a chamber” is three jobs			NOT ONE
STAGE	PRODUCES	COST / CATCH	
1 · Absorber unit cell FEM · Floquet · PML	α(f,θ) and complex surface impedance of the <i>real</i> wedge geometry	minutes · needs flow resistivity σ from an impedance tube	
2 · Interior field image source, or FEM/BEM/FDTD	p(x, f) with phase preserved, along each traverse	minutes → days · locally-reacting shortcut; wave solvers scale as f ³	
3 · Virtual qualification fit b − 20·log(ε/ε₀)	ΔLp vs the ISO tolerance table → a (frequency × distance × direction) envelope	free · must be driven with tones, not band noise	

Only stage 2 is expensive, and the published methodology (Bonfiglio & Pompoli) avoids meshing the room at all: it turns the stage-1 impedance into a spherical-wave oblique-incidence reflection coefficient and predicts the field with complex image sources. That is why chamber design is tractable on a laptop.

Method stack		ORDERED BY COST
METHOD	VALID WHERE — AND THE CATCH	
Equivalent fluid + TMM Delany-Bazley · Miki · JCA	α(f) of a flat layer from one parameter. Model choice moves the fitted σ by 2.5×	
FEM unit cell	α(f,θ) of any wedge or pyramid profile. Needs σ, which needs a measurement	
Complex image source	Interior field with phase, near-zero cost. Locally-reacting unless explicitly corrected	
Room FEM / BEM / FDTD	The only route that models extended-reaction absorbers. 6–10 DOF per wavelength; doubling f _{max} costs 8× the DOF	
Geometrical acoustics energy ray tracing	Carries energy, not phase → cannot represent cancellation → returns a clean, confident, wrong answer	

What the simulations get wrong

The input data is the error, not the solver. Uncertainty work targets three parameters: absorber flow resistivity, absorber thickness, hard-wall absorption. Vorländer: the input quality needed for JND-accurate prediction isn't available from reverberation-room measurements.

The diagnostic tell. When two numerical methods agree with each other but both miss the measurement, the material data is wrong — not the solver.

A model can only fail the way you asked it to. The Nash result is not a modelling artefact; it is a question-selection artefact. Reproduce a published benchmark case before trusting a first run.

The load-bearing conclusion. Nothing in this literature supports substituting a model for an ISO 26101-1 divergence-loss measurement. What the model buys is targeting — which frequencies, directions and distances are marginal, and therefore where the twenty measurements you can afford should go instead of the two hundred you cannot.

The workflow

1. Impedance tube (ISO 10534-2) on one flat sample of the lining — the only physical measurement stage A needs.
2. Fit σ with ≥3 models and keep the spread as an uncertainty band instead of picking a winner.
3. FEM the real geometry in a unit cell — one wedge, Floquet periodicity, PML. Output α(f,θ) and impedance.
4. Phase-preserving image-source model of the room with those coefficients; receivers on all eleven arc directions; include untreated features explicitly.
5. Run the virtual ISO 26101-1 test per direction; output the frequency × distance × direction map.
6. Excite it with tones at ½-octave centres. Tone-only failure is the result, not an artefact.
7. Then measure the flagged directions. Stage B — room FEM below the crossover with extended-reaction elements — only if stage A is marginal.

With no chamber at all: time-gate, planned with the image-source method. Gate length t_g sets the lowest trustworthy frequency at ≈ 1/t_g (5.6 ms → 178 Hz); image-source geometry gives the mounting height that buys it — for a 5 ms floor bounce at 0.48 m spacing, ≥ 1.08 m above the floor. Run any gating experiment with trigger.enabled = false.

Twelve ways a chamber lies to you

- 01 Cut-off is depth, not quality. Roughly where wedge length = λ/4. A better material in the same depth barely moves it.
- 02 Geometry carries the absorption. ITMO: a flat disk of the lining measures α < 0.5 (290–1800 Hz); the pyramid of the same material reaches 0.996.
- 03 Locally-reacting boundaries are wrong for porous wedges. Real absorbers are extended-reaction and angle-dependent — the standard shortcut, the standard optimism.
- 04 Your one input is ambiguous. Same sample, five models: σ = 3437 / 5136 / 5994 / 6125 / 8701 Pa·s/m².
- 05 Tones expose what noise hides. The Nash result. Decisive for tonal rotor noise.
- 06 A chamber is not anechoic — it is anechoic in an envelope. ITMO qualifies 50–3150 Hz at 0.5–0.9 m in every direction; past that the four directions diverge. Worst deviation: 15.5 dB.
- 07 Grates, ports, doors and recesses are the actual reflectors. ISO 26101-1 requires traverses aimed at exactly these features.
- 08 Your own rig is inside the field. Booms, arc frame, turntable — scatter that reads as directivity after the fact.
- 09 The fit changes the verdict. ISO 3745 fits a/(r−r₀) with a fitted acoustic-centre offset; ISO 26101 fits b − 20·log(r/r₀), r₀ = 1 m. Not always the same answer.
- 10 Small rooms can be unqualifiable by rule, not physics. Traverse minimum was λ/2 in ISO 26101:2012, λ/4 since 2017. Split's 2.8 × 1.7 × 2.05 m chamber bottomed out at 250 Hz.
- 11 Nobody ever tested the wedges. A purchased chamber is not a qualified one; one qualified once is not one qualified now.
- 12 A ray-tracing model will say it's fine. Round robins report up to ±5 dB max deviation on sound strength, worst where modal density is low.

What this means for the arc rig

ISO 5305 (rotor UAS < 150 kg) does not say “use a chamber”. It requires ISO 26101-1 inverse-square validity for *all source positions and measurement directions specified*, with traverses *along the measurement directions defined by the microphone positions*. For an 11-mic arc at 18° that is up to eleven traverse directions, not five.

Tolerance	±1.5 / ±1.0 / ±1.5 dB	125–630 · 800–5000 · ≥6300 Hz
Far field	R ≥ 5 · D _A	0.64 m for a 5" prop; 0.76 m for 6"
Ground clearance	H ₀ ≥ max(2 · D _A , 1.2 m)	aerodynamic rule; also sets floor-bounce path
Background noise	≥ 6 dB, prefer ≥ 15	failing bands must be stated, not silently included
Mic supports	acoustically treated	the arc frame is a scatterer in the field

The arc makes the room error worse, not neutral. Chamber deviation is direction-dependent. A single microphone samples one direction and picks up an offset; an elevation arc samples eleven simultaneously and plots them as a polar shape. A direction-dependent room artefact therefore lands on the *shape* of the directivity plot and looks exactly like real directivity — the same structural hazard as the recirculation finding, moved from the time axis to the spatial one. Comparisons at a *fixed* elevation stay far better protected, because the error is common-mode there.

Needed before stage 2 can run

- The arc radius. Not in config.toml, not in docs/mic_arc.md (which has all 11 serials and elevations). It decides whether the 275 stored takes are far-field at all.
- Room dimensions and surfaces — L × W × H, floor material, wall treatment, where the rig sits. The entire geometric input to stage A step 4.
- Rotor-plane height above the floor, and whether it can move. Prop diameters in the dp1 campaign, for the far-field and clearance numbers.
- Is a chamber actually on the table? Budget and space — or is this about characterising the room you already measure in? Stage A is identical either way; only what you do with the answer changes.

FULL REPORT & PRIMARY EVIDENCE

Full review — interactive, with figures
claude.ai/code/artifact/6226a13c-4e3e-43e6-bf94-a94fc8ca c625

Nash — Qualification of an anechoic chamber, ICA 2019
pub.dega-akustik.de/ICA2019/data/articles/001379.pdf

Bikmukhametov et al. — ITMO qualification envelope
arxiv.org/abs/2603.16556

STANDARDS & QUALIFICATION METHOD

ISO 26101-1:2021 — free-field qualification (preview)
cdn.standards.iteh.ai · ISO-26101-1-2021.pdf

ISO 5305:2024 — UAS noise measurement (preview)
cdn.standards.iteh.ai · ISO-5305-2024.pdf

Russo et al. — ISO 3745 vs ISO 26101, Euronoise 2018
euronoise2018.eu/docs/papers/369_Euronoise2018.pdf

BLOCKED — PAYWALLED, NEED THE PDFS

Claims resting on these are from abstracts, not full text; the first three are the core methodology papers. AIP, Elsevier and SAGE block automated retrieval and none has an open mirror — pull them through a proxy and drop them in papers/anechoic-simulation/.

Bonfiglio & Pompoli 2013 10.1121/1.4807820 · Schneider 2009 10.1016/j.jsv.2008.08.019 · Jiang et al. 2016 10.1016/j.jsv.2016.06.020 · Cunefare et al. 2003 10.1121/1.1527595 · Singh et al. 2020 10.1177/1475472X20905070 · Easw aran & Munjal 1993 10.1006/jsvi.1993.1027 · Wang & Tang 1996 10.1016/S0955-7997(96)00017-3 · Kar & Munjal 2006 10.1016/j.apacoust.2005.11.009 · Brinkmann et al. 2019 10.1121/1.5096178